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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP(HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEZGYA01 – FUNDAMENTALS OF SOIL ECOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Part A

Multiple Choice Question

Answer all questions

Each question carries **1** marks

1. **Actinorhizal plants are typically associated with:** [K] / [CO1, CO2, CO3, CO4, CO5]
   a) Legumes

2. **Mineralization is the process of :** [K]/ [CO2]
   b) Conversion of organic matter in soil into inorganic nutrients

3. **Biological interactions in soils mainly include:** [K]/ [CO3]
   a) Predation, competition, mutualism, parasitism

4. **Loamy soil is considered ideal for crops because:** [U]/ [CO5]
   b) It has balanced proportions of sand, silt, and clay

5. **Why are nematodes considered particularly indicative organisms for assessing the status of the soil?** [K] / [CO1, CO2, CO3, CO4, CO5]
   c) They are scarce

6. **Actinorhiza relationships are associated with fixation of:** [U] / [CO1, CO2, CO3, CO4, CO5]
   b) Nitrogen

7. **The litterbag method is used to measure:** [K]/ [CO2]
   b) Litter decomposition

8. **Anaerobic decomposition produces____.** [U]/ [CO2]
   a) CH_4

9. **Which fraction of soil organic matter is described as the "final product of decomposition" and contributes to soil structure and darkening of soil color?** [K] / [CO1, CO2, CO3, CO4]
   c) Humus

10. **The microbial conversion of organic nitrogen into ammonium is called_.** [U]/ [CO2]
    a) Ammonification

Part B

Short Answer

Answer any **14** questions

Each question carries **2** marks

11. **What are soil horizons?** [U]/ [CO5]

Distinct layers in a soil profile ordered from top to bottom: the organic (O), topsoil (A), eluviated (E), subsoil (B), parent material (C), and bedrock (R) horizons.

12. **Illustrate the link between decomposition and nutrient cycling?** [U]/ [CO2]

Decomposition is the process that directly links to nutrient cycling by breaking down dead organic matter which releases essential nutrients back into the soil. Soil organism consume dead plants and animals, converting complex organic compounds into simpler, inorganic form that plants can absorb and reuse.

13. **Explain how soils contribute to the global carbon cycle.** [U]/ [CO3]

By acting as the largest terrestrial carbon pool storing more carbon in the top 30 cm than the atmosphere and vegetation combined. Carbon is transferred to the soil via roots, leaves, and other debris. Soil organic matter (SOM) is the main component.

14. **Define megafauna and list two examples of megafauna.** [K]/ [CO3]

Soil megafauna are the largest soil-dwelling animals, typically these over 20 mm in length. Examples are: Moles, Gophers, Voles, Earthworm, larger Ants etc.

15. **Explain the role of soil microorganisms in humus formation.** [U]/ [CO2]

They decompose dead plant and animal matters, transform fresh residues into humus- breaking down materials like lignin and converting simpler by products into complex humus molecule.

16. **Why are soil microbes called "ecosystem engineers"?** [U] / [CO1, CO2, CO3, CO4, CO5]

They physically and chemically alter soil to create a healthy, stable environment for plants and other organism- they engineer soil structure, improves water infiltration and retension-responsible for vital nutrient cycling and help remediate pollution.

17. **Why is soil considered "dynamic"?** [K] / [CO1, CO2, CO3, CO4, CO5]

Because it is a complex living ecosystem constantly undergoing physical, chemical, and biological changes. Nutrient cycling, water and in movement; Living organisms and interactions; Management practices and long-term changes.

18. **What is common initial step when isolating microorganisms from a soil sample?** [K]/ [CO5]

To prepare a soil suspension by mixing a weighed amount of soil with a specific volume sterile diluents. It is the starting point for further isolation techniques.

19. **Discuss why mesofauna are important indicators of soil health.** [U]/ [CO3]

Their communities are sensitive to danger in the soil environment caused by human activities and natural disturbances, reflecting the overall health, fertility, and stability of the ecosystem can evaluate the impacts of land management practices on soil quality and ecosystem function (by monitoring density, diversity, and composition of mesofauna)

20. **Compare and contrast the feeding strategies of macrofauna detritivores (earthworms, millipedes) versus mesofauna detritivores (springtails, mites).** [U]/ [CO2]

Earthworm and millipedes are both crucial detrittivores and are essential members of the detritus food web. Millipedes primarily act as "shredders" of larger plant fragments, and earthworm functions more as "soil processors" that ingest and mix fine organic matter with soil as they burrow. Specific food preferences and feeding mechanism differ significantly.
Springtails and mites are vital components of soil ecosystem with a wide range of diets, but mites show greater diversity, including specialized parasitic and predatory roles. Both share detritivorous and some herbivorous tendencies. Mites roles as significant pests and disease vectors for both plants and animals.

21. **How do root exudates influence soil microbial activity?** [U] / [CO1, CO2, CO3, CO4, CO5]

By acting as a food source and signaling mechanism, which shapes the microbial community in the rhizosphere. These exudates provide nutrients that promote the growth of beneficial microbes then help plants with nutrient acquisition, growth, and stress tolerance. Nutrient source, attracting beneficial microbes, promoting plat growth, deterring pathogens altering soil properties and mediating signaling.

22. **How does lignin content affect the decomposition of plant materials?** [U]/ [CO2]

Higher lignin content slows down the decomposition of plant materials. Lignin is a complex recalcitrant polymer that is difficult for microorganism to breakdown: slows the rate, restricts access, requires specialized enzymes acts as a physical barrier, and contributes to recalcitrant pools.

23. **State the ecological role of root zone communities.** [K]/ [CO3]

Root zone communities are crucial for plant survival, nutrient cycling and soil stability. They acting as a complex interface between the plant and its environment. Their major roles are: nutrient and water uptake, soil structure and stability, nutrient cycling, microbial interactions, ecosystem stability and water regulation.

24. **Describe the structural difference between actinorhizal nodules and rhizobial nodules.** [K] / [CO1, CO2, CO3, CO4, CO5]

Actinorhizal nodules are modified lateral roots, while rhizobial nodules are modified system; like organs. Actinorhizal nodules leading to a less complex structure in contract rhizobial nodules are derived from the root cortex and pericycle developing a more complex, stem- like structure with a peripheral vascular system.

25. **Give an example of a soil insect and its ecological function.** [U]/ [CO3]

Examples: Springtails – acting as "cleaners" by breaking down dead organic matter and microbes.
Termites- ecosystem engineers.
Ants- soil aeration, nutrient cycling, pest control, and seed dispersal
Beetles- Decomposers, nutrient cyclers, and soil engineers
Other examples also may be included.

26. **Explain the role of mycorrhizal fungi in forest decomposition processes.** [U]/ [CO2]

By releasing enzymes that breakdown complex organic matter like dead leaves and wood. Key roles in decomposition are nutrient cycling, enzyme secretion, soil structure, carbon requestration, and network support.

Part C

Short Essay

Answer any **8** questions

Each question carries **4** marks

27. **Summarize the dynamics of litter breakdown.** [U]/ [CO2]

The dynamics a and litter breakdown stages are fragmentation, leaching, catabolism, humification, mineralization.
Major influencing factors are biotic (decomposers), abiotic (temperature and moisture, litter quality, soil properties)

28. **Write a note on symbiotic associations of roots with microbes.** [U] / [CO1, CO2, CO3, CO4, CO5]

It is a mutualistic relationship where plants and microorganisms such as fungi (mycorrhizae) and bacteria, exchange benefits like enhanced nutrient uptake, water absorption, and protection from pathogens.

Types of symbiotic associations are mycorrhizae (fungi-root association), rhizobia (bacteria- root association in legumes).

These symbiotic relationships allow plants to thrive, especially in poor soil conditions by making nutrients more accessible to plants.

29. **Describe characteristics of any five soil types.** [U]/ [CO5]

Sandy soil- weathered rock particles, large and loose, water drain quickly, holds very little moisture or nutrients.
Clay soil- very fine particles, dense and poor at draining water, sticky when wet and hard when dry
Silt soil- particles size are between clay and sand, smooth powdery texture, holds more moisture than sand.
Loam soil – best for gardening balanced mixture of sand, silt and clay along with organic matter; holds moisture well, while also allowing for good drainage.
Peat soil- dark and spongy formed from waterlogged around that has accumulated a high amount of organic matter.

30. **How does soil types affected decomposition?** [U]/ [CO2]

By influencing factors like moisture, aeration, and pH, which impact microbial activity. Sandy soils decompose slowly due to poor water and nutrient retention. Clay soils can protect organic matter from decomposition by forming aggregates that shield carbon stores from microbes. Compost manure – rich soils decompose organic matter very quickly because of their high moisture and nutrient content.

31. **Define field capacity in soils.** [K]/ [CO5]

It is the amount of water held in soil after excess water has drained away by gravity, typically 2-3 days after rainfall or irrigations. It represents the upper limit of plants available water where the soil is saturated but hot waterlogged. Factors like soil texture, structure, organic matter and temperature influence this value.

Field capacity is important for plant growth, irrigation management, and nutrient availability.

32. **Write a note on the role of protists in nutrient cycling.** [U] / [CO1, CO2, CO3, CO4, CO5]

By acting as decomposers that release inorganic nutrients from organic matter and as consumes that remineralize nutrients from prey. They play a crucial role in both soil and aquatic systems, contributing to the carbon and nitrogen cycles by breaking down organic matter and regulating microbial populations through predation. This process makes essential nutrients available for other organism such as plants, and influences overall ecosystem productivity.

Impacts of protists on soil and plant health includes rhizosphere and phyllosphere nitrogen and phosphorous and plant growth promotion.

33. **Explain the ecological role of nematodes and tardigrades in soil ecosystems.** [U]/ [CO3]

By participating in nutrient cycling, decomposition, and acting as indications of soil health. Prey ecological roles of nematodes in soil are: Nutrient cycling, decomposition, regulations of microbial communities, pest control, soil structure and indicators of soil health.

Tardigrades act as both consumers and prey, playing a crucial role in the good web by regulating populations of other microfauna like nematodes. Role as prey (food source inclusion in the food web); Role as consumers (predation on nematodes, population control, injuring prey). Major factors influencing their role are soil properties and community interactions.

34. **How do fauna influence microbial activity indirectly?** [U]/ [CO2]

Fauna indirectly influences microbial activity primarily through habitat modification organic matter processing, and nutrient cycling which change the physical and chemical environment in ways that favor specific microbial communities and processes. Key indirect mechanism include. Organic matter transformation, bioturbation and translocation, nutrient cycling and excretion, habitat alteration, dispersal of microorganisms and plant-microbe interations.

35. **Explain the process of primary production in soils with reference to energy capture.** [U] / [CO1, CO2, CO3, CO4, CO5]

Primary production in soils is primarily a function of plant photosynthesis, which captures solar energy and stores it as biomass, with a small contribution from chemoantotrophic bacteria using chemical energy. The captured energy enters the soil ecosystem mainly through plant roots and the decomposition of organic matter. The dominant energy capture mechanism is photosynthesis. The minor mechanism is chemosynthesis.

Primary production in soil is primarily an extension of above ground plant activity, relying on solar energy capture and its subsequent transfer below ground. The soil ecosystem acts as a critical hub for processing, cycling, and transferring this energy to support a vast and diverse community of organisms.

36. **Compare colonial growth forms and mycelia growth forms in terms of structure and function.** [U]/ [CO3]

Soil microorganisms primarily exhibit growth in one of two forms mycelia (filamentous) or colonial (unicellular/aggregates). These different structures dictate their unique roles in the soil ecosystem, particularly in nutrient cycling and soil structure formation.

Mycelial Growth Forms (Fungi and Actinomycetes) : Structure- filamentous network, large surface area, physical strength and long distance transport Function – decomposition of complex organic matter, nutrient cycling, symbiotic relationships (mycorrhizae), and disease suppression.

Colonial Growth Forms (Most Bacteria and Algae): Structure- unicellular or simple aggregates, high abundance, motility and bio film formation.

Function- fast nutrient cycling, nitrogen fixation and nitrification, mineral solubilization, response to disturbance, and pathogen consumptions.

The mycelia form provides structural stability and is crucial for breaking down tough organic materials and nutrient transport over large distances, while the colonial form excels in rapid nutrient cycling and diverse biochemical transformations in the soil microenvironment.